MODULE 32 · EXAM STRATEGY - ALL DOMAINS

Practice Exam 2 of 4

Twenty new questions, weighted like the real exam, including multiple-response questions.

QUESTIONS 20 PASS MARK 15 of 20 HOLD 9 seconds

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How to use this paper

- Pause after each question and commit to an answer before the solution appears.
- Write down your letters. Multiple-response questions need every correct letter.
- Score yourself by **domain** at the end, not only in total.
- For each miss, revisit the module named on the solution slide.

> ON THE EXAM
>
> The answer teaches the fact. The module teaches the discriminator you missed.

A V2 pipeline's deploy stage sometimes fails because a downstream API briefly throttles. Re-running the failed action always succeeds.

Engineers retry by hand several times a week. The team wants failed actions retried automatically, with no custom code.

What should the DevOps engineer configure?

A An On Failure stage condition with a Retry result

B A manual approval before the stage

C An EventBridge rule on failures invoking a Lambda function that retries

D Change the pipeline to QUEUED mode

◆ PAUSE AND DECIDE

What should the DevOps engineer configure?

A. An On Failure stage condition with a Retry result
Retries the failed actions automatically. Configuration only.

B. A manual approval before the stage
Adds a person; retries nothing.

C. An EventBridge rule on failures invoking a Lambda function that retries
Custom code for a built-in feature.

D. Change the pipeline to QUEUED mode
Changes ordering of runs, not retries.

REVISIT MODULE 2

Transient failure, same code: retry. Built into stage conditions since 2024.

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> Python builds in CodeBuild run in private subnets with no internet access. They need packages from the public PyPI registry.
>
> Security also wants a record and control of which package versions enter the company.

What should the DevOps engineer set up?

A. A NAT gateway so builds can reach PyPI directly

B. A nightly job that copies all of PyPI into an S3 bucket

C. A CodeArtifact repository with an external connection to PyPI, reached through CodeArtifact VPC endpoints

D. Developers download packages and commit them to the repository

◆ PAUSE AND DECIDE

What should the DevOps engineer set up?

A. A NAT gateway so builds can reach PyPI directly
Gives internet access, which the design forbids, and records nothing.

B. A nightly job that copies all of PyPI into an S3 bucket
Enormous, custom, and not a package repository.

C. A CodeArtifact repository with an external connection to PyPI, reached through CodeArtifact VPC endpoints
Packages are fetched and retained centrally; builds stay private.

D. Developers download packages and commit them to the repository
Manual, bloats the repository, and uncontrolled.

REVISIT MODULE 5

Private builds needing public packages: CodeArtifact external connection plus its endpoints. Python needs no S3 endpoint.

> A team deploys container images to ECS. Twice last month, a deployment picked up the wrong image because the latest tag had moved.
>
> The team wants every deployment traceable to an exact commit, with images that can never be silently replaced.

Which changes meet these requirements? (Choose THREE.)

- A Deploy using the latest tag, pulled at deployment time
- B Tag each image with the commit ID from the build
- C Enable tag immutability on the ECR repository
- D Expire all images older than one day
- E Pass the exact tag to the deploy action through an exported variable

◆ PAUSE AND DECIDE

Which changes meet these requirements? (Choose THREE.)

A Deploy using the latest tag, pulled at deployment time

The exact cause of the incidents.

B Tag each image with the commit ID from the build

Makes every image traceable to its source.

C Enable tag immutability on the ECR repository

A tag can never be pointed at a different image.

D Expire all images older than one day

Deletes images you may need to roll back to.

E Pass the exact tag to the deploy action through an exported variable

The deployment names the precise image the build produced.

REVISIT MODULE 5

Traceable, immutable, exact: commit tag, immutability, and passing that tag to the deploy.

> A shared Python library must be tested against Python 3.11, 3.12 and 3.13 on every commit, from a single CodeBuild project.
>
> The team wants the three test runs in parallel, with the least configuration.

What should the DevOps engineer use?

A Three separate CodeBuild projects

B A loop in the buildspec that installs each version in turn

C A Step Functions workflow starting three builds

D A batch build with a build matrix over the three versions

◆ PAUSE AND DECIDE

What should the DevOps engineer use?

A Three separate CodeBuild projects

Triples the configuration to maintain.

B A loop in the buildspec that installs each version in turn

Sequential, not parallel, and slower.

C A Step Functions workflow starting three builds

Custom orchestration for a built-in feature.

D A batch build with a build matrix over the three versions

One project; every combination runs in parallel.

REVISIT MODULE 3

Same project, several runtime versions, in parallel: a build matrix.

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A CDK pipeline adds a new target account. Deployment to it fails with an error that the bootstrap version parameter cannot be found.

The app deploys correctly to every other account.

What should the DevOps engineer do?

A Run cdk synth again and redeploy

B Bootstrap the new account and Region, trusting the pipeline account

C Grant the pipeline role administrator access in the new account

D Delete the cdk.out folder

◆ PAUSE AND DECIDE

What should the DevOps engineer do?

A Run cdk synth again and redeploy

The templates are fine; the account is not prepared.

B Bootstrap the new account and Region, trusting the pipeline account

Creates the bootstrap stack and roles the pipeline expects.

C Grant the pipeline role administrator access in the new account

There is still no bootstrap stack or role to assume.

D Delete the cdk.out folder

Local build output is not the problem.

REVISIT MODULE 11

Works everywhere except a new account, bootstrap missing: bootstrap it with --trust.

> Developers in 20 accounts need approved ElastiCache clusters with encryption and set node types.
>
> They must not hold ElastiCache permissions themselves. The platform team manages one catalogue centrally.

Which steps meet these requirements? (Choose TWO.)

A Share the Service Catalog portfolio with the developers' OU through Organizations

B Grant developers ElastiCache full access

C Deploy the cluster template with a StackSet

D Add a launch constraint role that can create the clusters

E An AWS Config rule for unencrypted clusters

◆ PAUSE AND DECIDE

Which steps meet these requirements? (Choose TWO.)

A Share the Service Catalog portfolio with the developers' OU through Organizations

One central catalogue, available in every account.

B Grant developers ElastiCache full access

The permission that is ruled out.

C Deploy the cluster template with a StackSet

Pushes clusters out; not self-service.

D Add a launch constraint role that can create the clusters

Service Catalog creates the cluster; developers need no ElastiCache rights.

E An AWS Config rule for unencrypted clusters

Detects afterwards; enables nothing.

REVISIT MODULE 10

Central catalogue in many accounts, no underlying permissions: share the portfolio, plus a launch constraint.

> Engineers sometimes edit CloudFormation-managed resources by hand in 60 accounts.
>
> The compliance team wants stacks with drift detected daily and reported centrally, with no custom code.

What should the DevOps engineer set up?

A. A scheduled Lambda function in each account calling drift detection

B. Drift-aware change sets on every stack

C. The cloudformation-stack-drift-detection-check Config rule through an organisation conformance pack, with an aggregator

D. CloudTrail Insights

◆ PAUSE AND DECIDE

What should the DevOps engineer set up?

A. A scheduled Lambda function in each account calling drift detection
Custom code in sixty accounts.

B. Drift-aware change sets on every stack
Restores drift during a deployment; does not detect and report it daily.

C. The cloudformation-stack-drift-detection-check Config rule through an organisation conformance pack, with an aggregator
Managed detection everywhere, reported in one place.

D. CloudTrail Insights
Flags unusual API volumes, not drift.

REVISIT MODULE 9

Detect and report drift everywhere: the managed Config rule. Restore it on deploy: drift-aware change sets.

A Systems Manager Automation runbook must run in 50 accounts and three Regions to update a configuration value.

The operations team works from a delegated administrator account and wants one execution with rate control.

What should the DevOps engineer do?

A Start a multi-account, multi-Region Automation execution targeting the OUs and Regions

B A script that loops through each account and Region

C A StackSet with a custom resource that runs the change

D Run Command separately in each account

◆ PAUSE AND DECIDE

What should the DevOps engineer do?

A Start a multi-account, multi-Region Automation execution targeting the OUs and Regions

One execution, fanned out with concurrency and error limits.

B A script that loops through each account and Region

Custom code doing what Automation does natively.

C A StackSet with a custom resource that runs the change

Misuses infrastructure deployment for an operation.

D Run Command separately in each account

One hundred and fifty separate executions.

REVISIT MODULE 13

Same runbook everywhere, one execution, rate control: multi-account, multi-Region Automation.

A Lambda function consumes an SQS queue and writes to a database that handles only 20 concurrent writers.

During bursts, Lambda scales up and overloads the database. The team wants to cap it without throttling errors or extra retries.

What should the DevOps engineer configure?

A Reserved concurrency of 20 on the function

B A batch size of 1

C A shorter visibility timeout

D Maximum concurrency of 20 on the SQS event source mapping

◆ PAUSE AND DECIDE

What should the DevOps engineer configure?

A Reserved concurrency of 20 on the function

Caps it, but pollers keep delivering and get throttled, causing retries.

B A batch size of 1

More invocations, not fewer.

C A shorter visibility timeout

Causes duplicate processing.

D Maximum concurrency of 20 on the SQS event source mapping

Pollers stop at 20, so nothing is throttled.

REVISIT MODULE 15

Cap a queue consumer cleanly: maximum concurrency on the event source mapping, not reserved concurrency.

> An Aurora PostgreSQL cluster serves read-heavy reporting whose volume swings unpredictably through the day.
>
> The team wants reader capacity added and removed automatically, with no manual steps.

What should the DevOps engineer configure?

A Add Aurora replicas by hand each morning

B An Aurora Auto Scaling policy for replicas, tracking reader CPU

C Convert to a Multi-AZ DB instance deployment

D Convert to an Aurora Global Database

◆ PAUSE AND DECIDE

What should the DevOps engineer configure?

A Add Aurora replicas by hand each morning
Manual, and the swings are unpredictable.

B An Aurora Auto Scaling policy for replicas, tracking reader CPU
Adds and removes replicas as load changes.

C Convert to a Multi-AZ DB instance deployment
For failover; its standby serves no reads.

D Convert to an Aurora Global Database
Cross-Region protection, not read scaling.

REVISIT MODULE 15

Unpredictable read load: replica auto scaling with target tracking.

> An API runs in two Regions, active-passive. The primary can be up while its database connection is broken.
>
> Traffic must move to the secondary automatically whenever the primary cannot serve real requests.

Which steps meet these requirements? (Choose TWO.)

A Simple routing to the primary Region

B Failover routing records, with a health check on the primary

C A one-day TTL on the records

D Weighted routing, 50 to each Region

E Point the health check at a deep endpoint that verifies the database

◆ PAUSE AND DECIDE

Which steps meet these requirements? (Choose TWO.)

A Simple routing to the primary Region

Simple routing has no failover.

B Failover routing records, with a health check on the primary

Moves traffic when the primary is unhealthy.

C A one-day TTL on the records

Slows failover for a day.

D Weighted routing, 50 to each Region

Active-active, not failover.

E Point the health check at a deep endpoint that verifies the database

The check then fails when real requests would.

REVISIT MODULE 14

Failover needs a routing policy and a check that tells the truth.

A hundred Lambda functions in production write large volumes of debug logs through the runtime's standard logger. Ingestion costs are high.

The team wants debug lines dropped before ingestion, with no code changes.

What should the DevOps engineer do?

A. Set JSON log format and an application log level of WARN in each function's logging configuration

B. Switch the log groups to the Infrequent Access class

C. A subscription filter that drops debug lines

D. Shorten retention to one day

◆ PAUSE AND DECIDE

What should the DevOps engineer do?

A Set JSON log format and an application log level of WARN in each function's logging configuration

Lambda filters lower-level lines before they are sent. No code change.

B Switch the log groups to the Infrequent Access class

Cheaper per GB, but the debug lines are still ingested.

C A subscription filter that drops debug lines

Subscriptions copy logs after ingestion; they drop nothing.

D Shorten retention to one day

Storage cost falls; ingestion cost does not.

REVISIT MODULE 17

Stop paying to ingest noise, no code change: Lambda's log level controls.

> The team needs an alarm on the fault rate of one API path, /checkout, which is traced with X-Ray.
>
> Other paths should not affect the alarm.

What should the DevOps engineer do?

A. A Contributor Insights rule on the application logs

B. A CloudWatch Synthetics canary on /checkout

C. An X-Ray group with a filter expression for /checkout, and an alarm on its fault metric

D. A metric filter on the load balancer access logs

◆ PAUSE AND DECIDE

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What should the DevOps engineer do?

A A Contributor Insights rule on the application logs

Ranks top contributors; not a fault-rate metric for one path.

B A CloudWatch Synthetics canary on /checkout

Tests synthetic requests, not real traffic's fault rate.

C An X-Ray group with a filter expression for /checkout, and an alarm on its fault metric

Groups publish metrics for only the matching traces.

D A metric filter on the load balancer access logs

Access logs go to S3, where metric filters do not apply.

REVISIT MODULE 19

One path's traced fault rate: an X-Ray group, which publishes its own metrics.

> Analysts run the same Logs Insights queries many times a day, each filtering a huge log group by requestId.
>
> Queries are slow and scan costs are rising.

What should the DevOps engineer do?

A Increase the log group's retention

B Switch the log group to the Infrequent Access class

C Export the logs to S3 every day

D Create a field index on requestId

◆ PAUSE AND DECIDE

What should the DevOps engineer do?

A Increase the log group's retention

More data, not less scanning.

B Switch the log group to the Infrequent Access class

Changes ingestion price, not query scanning.

C Export the logs to S3 every day

Moves the problem and adds a pipeline.

D Create a field index on requestId

Filtered queries scan only matching events: faster and cheaper.

REVISIT MODULE 18

Repeated filters on one field: a field index cuts what each query scans.

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> AWS schedules EC2 instances for retirement across 40 accounts. Each notice has a deadline, and missing it means an unplanned stop.
>
> The company wants each affected instance stopped and started automatically, at a quiet time before the deadline.

What should the DevOps engineer do?

A Email the notices to the owning teams

B An EventBridge rule on AWS Health retirement events that starts a stop-and-start Automation runbook

C An alarm that reboots the instances

D An AWS Config rule for scheduled retirements

◆ PAUSE AND DECIDE

What should the DevOps engineer do?

A Email the notices to the owning teams

Manual, and easy to miss.

B An EventBridge rule on AWS Health retirement events that starts a stop-and-start Automation runbook

Acts on every notice automatically; stop and start moves the instance to new hardware.

C An alarm that reboots the instances

A reboot stays on the same host.

D An AWS Config rule for scheduled retirements

Config does not see Health events.

REVISIT MODULE 21

Announced failures: AWS Health event, rule, runbook. Stop and start moves hosts; reboot does not.

> When any production pipeline fails, the on-call team wants a message in its Slack channel and an OpsItem opened for tracking.
>
> No custom code is allowed.

Which steps meet these requirements? (Choose TWO.)

A. An EventBridge rule on failed pipeline executions targeting an SNS topic mapped in Q Developer in chat applications

B. A Lambda function that polls pipeline status every minute

C. The same EventBridge rule, also targeting OpsCenter to create an OpsItem

D. CloudTrail data events for CodePipeline

E. Email the Slack channel's address

◆ PAUSE AND DECIDE

Which steps meet these requirements? (Choose TWO.)

A An EventBridge rule on failed pipeline executions targeting an SNS topic mapped in Q Developer in chat applications

Formatted Slack messages, no code.

B A Lambda function that polls pipeline status every minute

Custom code, and polling.

C The same EventBridge rule, also targeting OpsCenter to create an OpsItem

EventBridge can create OpsItems directly.

D CloudTrail data events for CodePipeline

Records API calls; notifies nobody.

E Email the Slack channel's address

Unformatted, and no OpsItem.

REVISIT MODULE 22

One event, two targets: chat notification and an OpsItem. No code needed.

> A new ECS service on Fargate fails to start. The stopped reason is CannotPullContainerError with an authorisation failure from ECR.
>
> The image exists, and the task role has full ECR permissions.

What should the DevOps engineer fix?

A Add more ECR permissions to the task role

B Enable privileged mode

C Increase the task's memory

D Give the task execution role ECR pull permissions, such as the AmazonECSTaskExecutionRolePolicy

◆ PAUSE AND DECIDE

What should the DevOps engineer fix?

A Add more ECR permissions to the task role

The task role is used by the application, after the image is pulled.

B Enable privileged mode

Not available on Fargate, and unrelated.

C Increase the task's memory

The task never started.

D Give the task execution role ECR pull permissions, such as the AmazonECSTaskExecutionRolePolicy

ECS pulls images with the execution role, before your code runs.

REVISIT MODULE 23

Fails before the code runs: execution role. The task role already had permissions: that is the planted detail.

> A developer's role has an identity policy allowing dynamodb:*. A call to CreateTable fails with an explicit deny in a permissions boundary.
>
> The developer asks for the identity policy to be widened.

What should the DevOps engineer do?

A Widen the identity policy further

B Add an allow to an SCP

C Review the permissions boundary, and allow CreateTable there if it is appropriate

D Add a resource-based policy to the table

◆ PAUSE AND DECIDE

What should the DevOps engineer do?

A Widen the identity policy further

It already allows the action. The boundary caps it.

B Add an allow to an SCP

The message names a boundary, not an SCP.

C Review the permissions boundary, and allow CreateTable there if it is appropriate

The boundary is the ceiling the message names.

D Add a resource-based policy to the table

The table does not exist yet, and the boundary still caps it.

REVISIT MODULE 24

Read the policy type in the message. A boundary deny is a ceiling, not a missing grant.

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> An audit finds customer managed KMS keys whose key material has not changed in years.
>
> Policy now requires key material to rotate automatically at least yearly, without re-encrypting stored data.

What should the DevOps engineer do?

A Create a new key each year and re-encrypt all data

B Enable automatic rotation on the keys

C Switch to AWS managed keys

D Import new key material each year

◆ PAUSE AND DECIDE

What should the DevOps engineer do?

A Create a new key each year and re-encrypt all data

Re-encryption is what the policy wants to avoid.

B Enable automatic rotation on the keys

New material on schedule; old material kept, so old data still decrypts.

C Switch to AWS managed keys

Loses control of the key policy, and cross-account use.

D Import new key material each year

Imported material cannot rotate automatically.

REVISIT MODULE 25

Rotation keeps old material: nothing needs re-encrypting.

> A shared S3 bucket must be readable by any principal in the company's organisation, currently 300 accounts, and by nobody else.
>
> New accounts join every week.

What should the DevOps engineer put in the bucket policy?

A An allow for s3:GetObject conditioned on aws:PrincipalOrgID matching the organisation

B A list of all 300 account ARNs

C Make the bucket public and restrict it with an SCP

D Share the bucket through AWS RAM

◆ PAUSE AND DECIDE

What should the DevOps engineer put in the bucket policy?

A An allow for s3:GetObject conditioned on aws:PrincipalOrgID matching the organisation
Covers every current and future account in one condition.

B A list of all 300 account ARNs
Needs editing every week, and grows unwieldy.

C Make the bucket public and restrict it with an SCP
SCPs cannot stop outsiders reading a public bucket.

D Share the bucket through AWS RAM
S3 buckets are not shared through RAM.

REVISIT MODULE 26

'Our organisation, including new accounts': aws:PrincipalOrgID.

Answer key

- **Answer key:** 1 A, 2 C, 3 BCE, 4 D, 5 B, 6 AD, 7 C, 8 A, 9 D, 10 B
- **Answer key:** 11 BE, 12 A, 13 C, 14 D, 15 B, 16 AC, 17 D, 18 C, 19 B, 20 A
- Multiple response (3, 6, 11, 16) score only if every letter is right.
- 15 or more: on track. 13 or 14: close. 12 or fewer: revisit weak domains first.

> ON THE EXAM
>
> Each solution slide named the module to revisit. Use it for every miss.

Scoring by domain

DOMAIN	QUESTIONS	MODULES TO REVISIT
D1 SDLC Automation	1, 2, 3, 4	2, 3, 5
D2 Configuration Management and IaC	5, 6, 7, 8	9, 10, 11, 13
D3 Resilient Cloud Solutions	9, 10, 11	14, 15
D4 Monitoring and Logging	12, 13, 14	17, 18, 19
D5 Incident and Event Response	15, 16, 17	21, 22, 23
D6 Security and Compliance	18, 19, 20	24, 25, 26

ON THE EXAM

Two misses in one domain matter more than three spread across six.